Mastering Distributed System Debugging: A Guide for Modern Engineers



Traditional distributed system debugging done using core visibility tools is going through massive changes with the introduction of AI.

Debugging distributed systems is probably one of the hardest jobs in software engineering today but is also one of the most satisfying. Once your application starts running across multiple microservices, Kubernetes clusters, cloud regions, and external dependencies, troubleshooting becomes very different from traditional debugging.

Earlier, engineers mostly checked logs from one machine or one process. Modern systems do not work like that anymore. One request can travel through ten or twenty services before the response comes back. If one dependency slows down or fails unexpectedly, the entire chain gets affected. Because of this, debugging is no longer just about reading logs. It requires visibility into the complete system behaviour.

The shift in reliability culture

A few years ago, developers and operations teams worked separately in many organisations. Developers focused on writing features while operations teams handled production support. In practice, this separation created many problems because the people writing the code were often disconnected from production behaviour.